

TECHNICAL OVERVIEW

Processor switching: All system resources are available to both the Z80 and the 68000 but in real time, only one processor is active at a time. Virtual co-processing is possible. The board's bus structure is essentially a 16 bit data path supported by an effective 24 bit address bus plus control lines. Switching from one processor to another is accomplished by executing one I/O instruction (Z80) or one memory reference instruction (68000) at the proper port or memory address location. Switching is accomplished through bus arbitration logic. Bus arbitration meant in the sense that the 68000 grants system resources (the bus) to the Z80 through the BR*,BG*,BGACK* 68000 control lines while the 68000 grants control to the Z80 via the Z80's BUSRQ* AND BUSAK* lines.

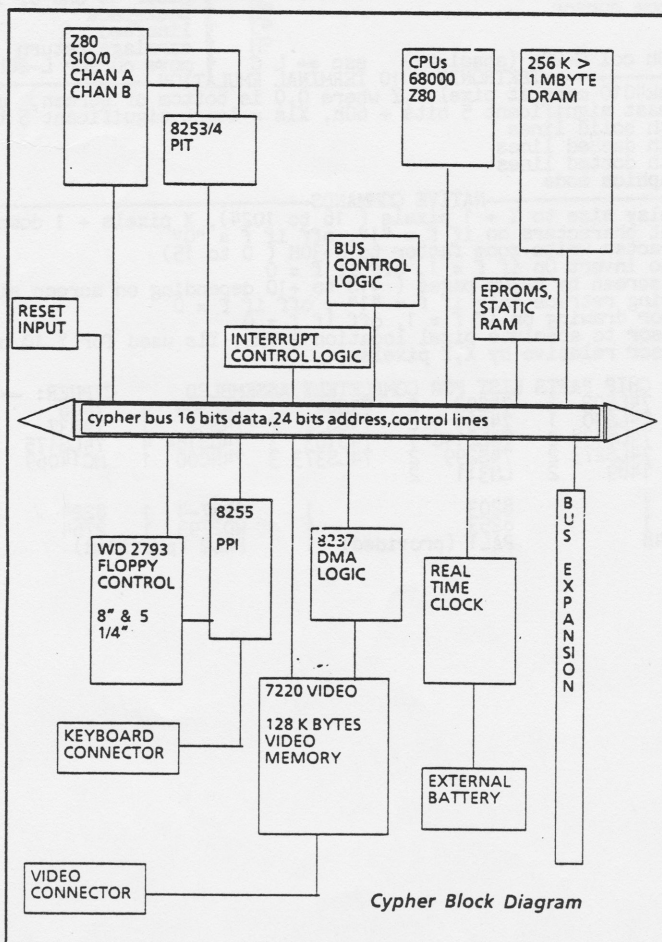
Memory resources: There are several types of memory available in the system. There are two 28 pin EPROM sockets which can therefore support a non-volatile R/W storage from 16K Bytes (2764 EPROMS) to 64K Bytes (27256 EPROMS). Primary R/W storage is implemented in Dynamic Ram with total storage ranging from 256K Bytes (4164 Dynamic RAMS) to One M Bytes (41256 Dynamic RAMS). Additional static ram is available via two 28 pin EPROM/SDRAM compatible sockets. A small amount (50 Bytes) of non-volatile R/W storage is available in the real-time clock section. In addition there are 128K Bytes of R/W storage available in the graphics logic section. The graphics memory normally is used for display but could be used in whole or in part to implement an auxiliary R/W storage element (RAM DISK).

All I/O devices with exception of the 12 bit D/A, A/D ports are 8 bit ports which reside at ODD addresses within the system. The memory and I/O map are dependent upon which processor is active although they are nearly the same for each processor. Each processor has read/write memory extending from address 000000H to address 0FFFFFFH (1 Mega-byte on board). Static ram appears at address FE0000H to address FFFFFFFH for the 68000 and at address XXF000H to address XXXFFFFH for the Z80. The static ram 'window' for the Z80 is essential for its paging of memory in the implementation of operating systems such as CP/M PLUS. The static ram for the 68000 only appears at address FE0000H to FFFFFFFH. The eprom memory resides from address FE0000H to FFFFFFFH. The Z80 selects it's current 60K page of memory by writing the page address to an eight bit port. This port supplies the upper eight address lines of the 24 bit address bus during Z80 memory cycles and 8237 DMA memory cycles. The non-directly addressable system storage of the VIDEO memory is accessed by the NEC 7220 through an eight bit I/O port. Transfers can be done by processor control or through a 8237 DMA channel. The non-volatile and non-directly addressable memory of the real time clock is accessed also as an I/O port.

Memory Map at Boot up: System boot up is done under Z80 control. At power up or reset the EPROM memory appears by default at address zero. The Z80 moves code to the static ram window, enables the dynamic ram memory at address zero, and then writes the 68000's 1K byte exception vector table. The Z80 then terminates execution (if a 68000 is present) and waits for the 68000 to come out of reset (the 68000 reset time is greater than the Z80's) and transfers control to it when appropriate.

System I/O: The Z80 addresses system I/O devices as I/O ports. The 68000 addresses I/O devices as memory locations. The top 4 K bytes of the 68000 address space generates I/O control signals which mimic the manner in which the Z80 accesses devices. To the programmer however, the I/O devices are simply memory locations. Either processor can access and control all I/O devices. A systems programmer may find it convenient to delegate most or all I/O driver routines to the eight bit Z80. This approach eliminates the need to write duplicate driver routines for each processor. However it might be more advantageous in certain circumstances to have the 68000 do all I/O. This could be useful when implementing multipage Z80 operating systems (CP/M PLUS) since the 68000's linear addressing capability could move data efficiently across 64K byte page boundaries for the Z80. A simple way of having the Z80 do I/O tasks for the 68000 is to allot an inter-processor communications (IPC) area of memory. When the 68000 requires a certain I/O task performed (such as a disk read) a code is written into the IPC. The 68000 then requests Z80 processing, by writing to the processor swap address (0FFFFFFH for the 68000). The Z80 starts processing, examines the IPC, carries out the proper task, writes to port OFF (its processor swap port), and the 68000 then continues operation. Proper error flags as set by the Z80 would be examined.

Functional Blocks: The Cypher Computer, like most other computers, consists of a CPU section, memory section, I/O section and graphics section. The block diagram of the system is shown below. The data Bus of the Cypher is 16 bits. When the Z80 is using the bus it reads/writes even addresses from the upper 8 data bits ie. D8-D15. When the Z80 is accessing an odd address it reads/writes data from the lower 8 data bits ie. D0-D7. This scheme applies to all memory and peripheral devices. All peripherals except the 12 bit D/A and A/D section are eight bits and are accessed on the lower 8 data bits ie. D0-D7. It must be remembered that when accessing I/O devices that they always appear at an ODD address. The 68000 has a 4 K byte section of memory reserved from address 0FF800H to 0FFFFFFH for I/O transactions. For example Serial I/O Channel A Data Port appears at Address 0FF8B1H. The Z80 accesses the same Data Channel at address 0B1H. The DMA channel addressing is slightly more complex for the Z80 in that it must use the Z80 instruction OUT (C),A in order to achieve a 16 bit I/O address for the 8237 decoding.



Cypher Block Diagram

FIRMWARE SECTION:

Our Firmware is constantly being enhanced, the following is a summary of our latest firmware.

68000 Monitor: THE FOLLOWING IS DISPLAYED BY THE 68K MONITOR IF AN INVALID COMMAND IS ENTERED.

```
COMMAND GET:          syntax
D(UMP)               D<START>,<END>
M(MEMORY)            M<ADDRESS>
F(ILL)               F<START>,<END>,<BYTE>
C(COPY)              C<SOURCE_START>,<SOURCE_END>,<DEST_START>
V(ERIFY)             V<SOURCE_START>,<SOURCE_END>,<DEST_START>
G(OTO)               G<ADDRESS>
S(WITCH)             S (switch to z80 monitor)
A(CCEPT)           A<OPTIONAL OFFSET TO LOAD ADDRESS>
R(EGISTER)           R<REG IDENTIFIER> <VALUE> (all reg. display)
```

(Note: CPM-80 OR CYPHER-DOS is used to boot in CPM-68k)

Z80 Monitor: THE FOLLOWING MENU IS DISPLAYED BY THE Z80 MONITOR IF AN INVALID COMMAND IS ENTERED.

```
COMMAND SET:      syntax
D(UMP)           D<START> <END>
M(MEMORY)        M<ADDRESS>
F(ILL)           F<START>, <END>, <BYTE>
C(COPY)          C<SOURCE START>, <SOURCE END>, <DEST START>
V(VERIFY)        V<SOURCE START>, <SOURCE END>, <DEST START>
T(EST)           T<MEMORY START>, <MEMORY END>, <DRAM BANK>
G(GOTO)          G<ADDRESS>
I(INPUT)         I<PORT#>
O(OUTPUT)        O<PORT#>, <DATA>
S(SWITCH)        S (switch to 68000 CPU)
X(SELECT)         X<8-3" 5-5", <0.1 -Side>, <1-S 2-D Density>
Y(DMA)           Y<DMA READ/WRITE ADDRESS> FE00 DEFAULT
Z(SECTOR)        Z<SIZE> 80 HEX DEFAULT
R(EAD)           R<UNIT>, <TRACK>, <SECTOR>
W(RITE)          W<UNIT>, <TRACK>, <SECTOR>
B(BOOT)          B<DRAM BANK#> (boot up cpm or CYPHER-DQS)
```

The Z80 and 68000 monitors have various inter-processor slave commands that allows one CPU to give tasks or commands for the other CPU to execute, i.e. Z80 commands the 68000 to do a memory block move, or the 68000 commands the Z80 to do disk I/O.

Textronix and ADM 3A terminal emulation functions on the Nec 7220 128k byte graphics provided in EPROM:

Value	Function	Value	Function
-----ADM 3A TERMINAL EMULATION-----			
ESC	move cursor up	ESC	advanced cursor right
ESC	home cursor	ESC	clear to end of line
ESC	clear screen and home cursor	ESC	backspace
ESC	tab	ESC	linefeed
ESC	home cursor	ESC	carriage return
esc = L C	move cursor to L-20h col C-20h (absolute)	esc =+ L C	move cursor L-20h lines, C-20h cols. (relative)
-----TEKTRONIX 4010 TERMINAL EMULATION-----			
esc GS Yms Yls Xms Xls	enter Tek4010 mode at pixel X,Y where 0,0 is bottom of screen, m=most significant 5 bits +20h Yls = least significant 5 bits + 60h, Xls = least significant 5 bits of X + 40h		
esc s	draw with solid lines		
esc b	draw with dashed lines		
esc d	draw with dotted lines		
esc FF	exit graphics mode		
-----NATIVE COMMANDS-----			
esc R Yms Yls Xms Xls	set display size to Y + 1 pixels (16 to 1024), X pixels + 1 downs (1 to 1024)		
esc f	set slant characters on if f = "1", off if f = "0"		
esc Z n	set character write zoom factor to n-30h (0 to 15)		
esc v f	set video invert On if f = 1, off if f = 0		
esc P n	advance screen by n-30h pages (-10 to +10 depending on screen size, screen paging)		
esc t f	draw during retrace only if f = "1" off if f = 0		
esc C f	set cursor drawing on if f = 1, off if f = 0		
esc A Yms Yls Xms Xls	move cursor to absolute pixel location (Yms, Yls used for Y 10 bit address)		
esc B Yms Yls Xms Xls	move cursor relative by X,Y pixels		

CHIP PARTS				LIST FOR COMPLETELY ASSEMBLED				CYPPER:							
74LS00	3	74S00	1	74LS01	1	74LS02	3	74S04	1	74LS04	4	74LS08	4	74LS11	1
74LS14	1	74LS20	3	74LS22	1	74LS30	3	74LS32	7	74LS37	1	74LS74	6	74LS74	3
74LS86	1	74LS123	1	74LS133	1	74LS138	2	74LS163	3	74LS164	4	74LS175	1	74LS240	3
74LS245	10	74LS257	6	74LS257	1	74LS273	2	74LS299	2	74LS373	3	74HC00	1	MC14069	1
AD7545	2	555	1	1483	2	1489	2	LM311	2						
MC68000L8	1	Z80A			1	8203			1	8237-4	1	8224		8255A	1
NEC7220	1	Z80A SIO/O				8253				WD2793	1	2764		2(provided)	
0116	2	41256		48		PAL1 (provided)			1	PAL2 (provided)					